AgentDebate

USER GUIDE · LOCAL EDITION

From a question to an analyst decision

Agent Debate & Panel Room

Prepare → **Debate** → **Verify** → **Act**

Workflows, illustrated screens, evidence review, reporting, and implementation reference.

22 October 2026
Based on the AgentDebate-local application

> Screenshots show the actual interface with isolated demonstration records. They are instructional examples, not real operational findings. No live model requests or changes to your saved analysis were needed to create this guide.

How to use this guide

Use the analyst chapters to run a session. Use the implementation chapters to operate and understand the local application.

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The example used throughout

A regional service must recover after a power interruption. The analyst is evaluating whether to rely on backup capacity or a failover procedure. An exercise showed a delay; the cause and peak-load capacity remain uncertain.

The useful outcome is a defensible disposition, an evidence gap, and an observable next action—not merely agreement among simulated SMEs.

> SMEs are LLM-generated perspectives based on saved profiles. The analyst remains responsible for evidence judgments, decisions, and follow-up.

One application, two debate workflows

Choose	Best suited to	How it runs
Agent Debate	Risk discovery, challenging a plan, or comparing explicit options against acceptance criteria.	A Node server worker executes the guided analysis and saves progress.
Panel Room	Exploring disagreement through structured phases, or facilitating an open discussion.	Browser-driven turns call the local server. Automatic structured debate runs to analyst resolution.

The shared workflow

Scenario + evidence → **Question + SMEs** → **Saved debate**

Session Tracker → **Summary / Reports** → **Follow-up**

Where to go

Sidebar area	Purpose
Prepare	Scenarios, Decisions, and Knowledge Base supply reusable context.
Run	Agent Debate starts guided analysis; Panel Room starts facilitated or automatic discussion.
Manage	Session Tracker manages analyst review; Session Summary is the reference view; Reports produces shareable outputs.
Library	SMEs, Domains, Threats, and Chains manage reusable perspectives and linked analytical records.

Important distinctions

A **scenario** describes the situation. A **session** is one debate about that situation. A **finding** is a claim to assess. A **mitigation or follow-up** is work to perform and verify. Multiple sessions can examine the same scenario; always check the question, mode, and date.

Prepare a scenario and evidence

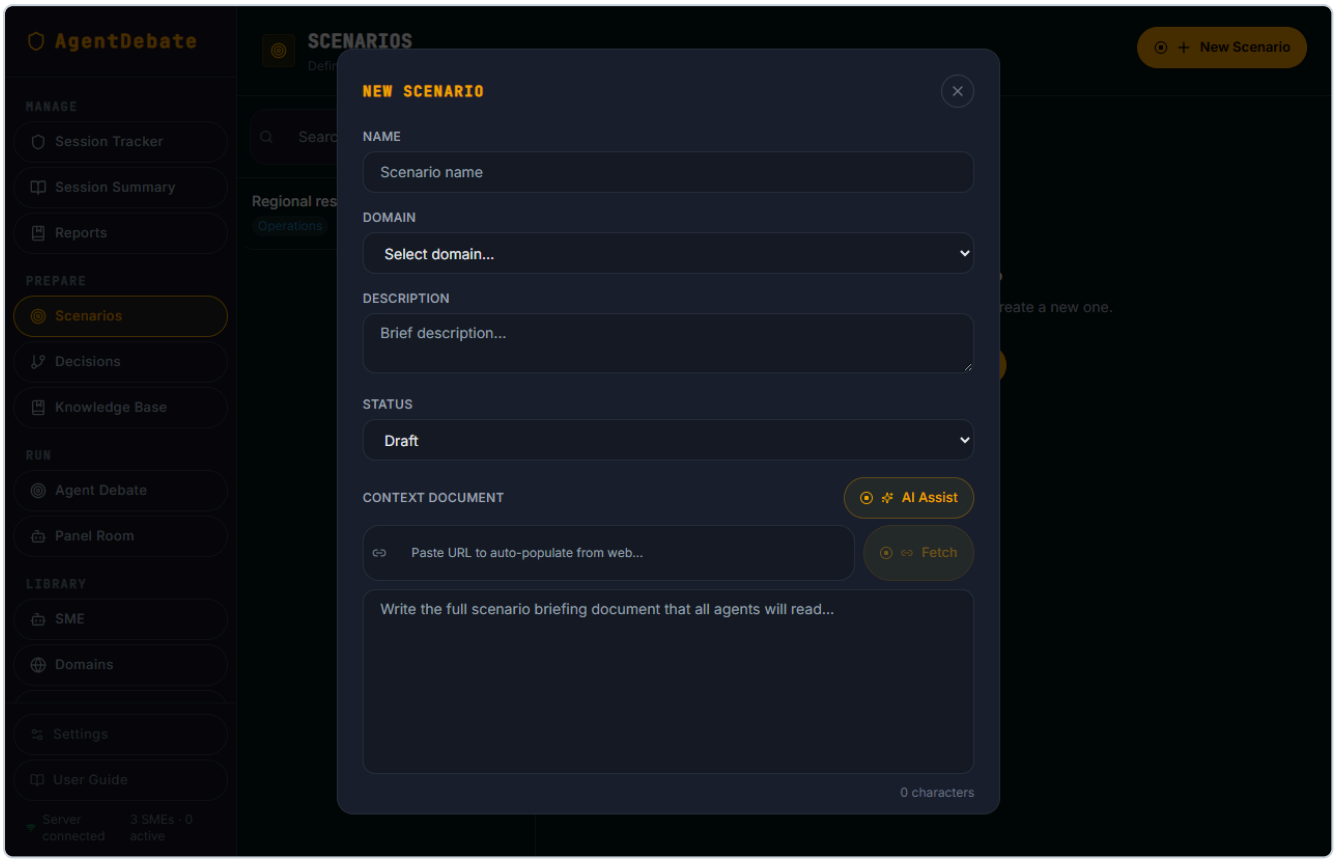


Figure 1. Scenarios are reusable context for subsequent sessions. · Demonstration data

1. Open **Prepare** → **Scenarios**. Choose or create the scenario that describes your problem.
2. Include the environment, stakeholders, known facts, sources, and uncertainties. A title alone does not provide enough context.
3. Use **Knowledge Base** for source material. Identify source dates and limitations; distinguish supplied evidence from assumptions.
4. For option comparison, prepare a **Decision** with distinct options and acceptance criteria.

> **Example question:** Which recovery approach can meet the target under peak demand, and what evidence would change that choice?
> **Example evidence gap:** No current peak-load test establishes backup capacity.

Select relevant SME perspectives

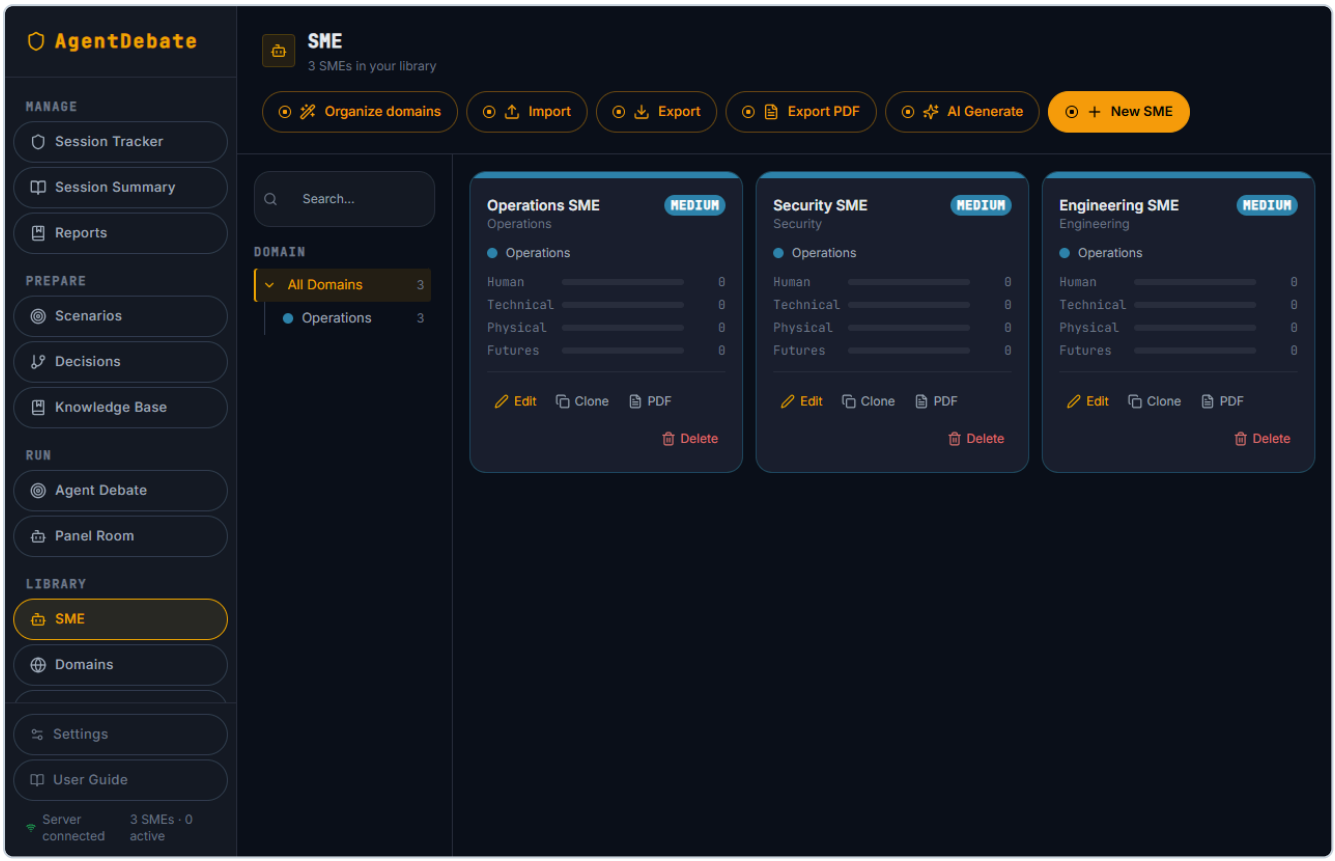


Figure 2. SME profiles and domain organization support panel selection. · Demonstration data

Domains organize expertise; domain colors help recognize related profiles. Empty domains do not themselves contribute an SME perspective. Review the actual panel rather than assuming the domain tree guarantees coverage.

1. Check SME descriptions and expertise against the question. For the example, include operations, security, and engineering perspectives.
2. Review the suggested panel: suggestions use topic matching and discipline diversity, not an independent model assessment of suitability.
3. Keep a panel of 3–8 suitable SMEs. Remove irrelevant profiles and add missing perspectives before running.

> More SMEs mean more model work. Diverse labels do not guarantee independent evidence: every contribution must still be assessed against the supplied record.

Run Agent Debate

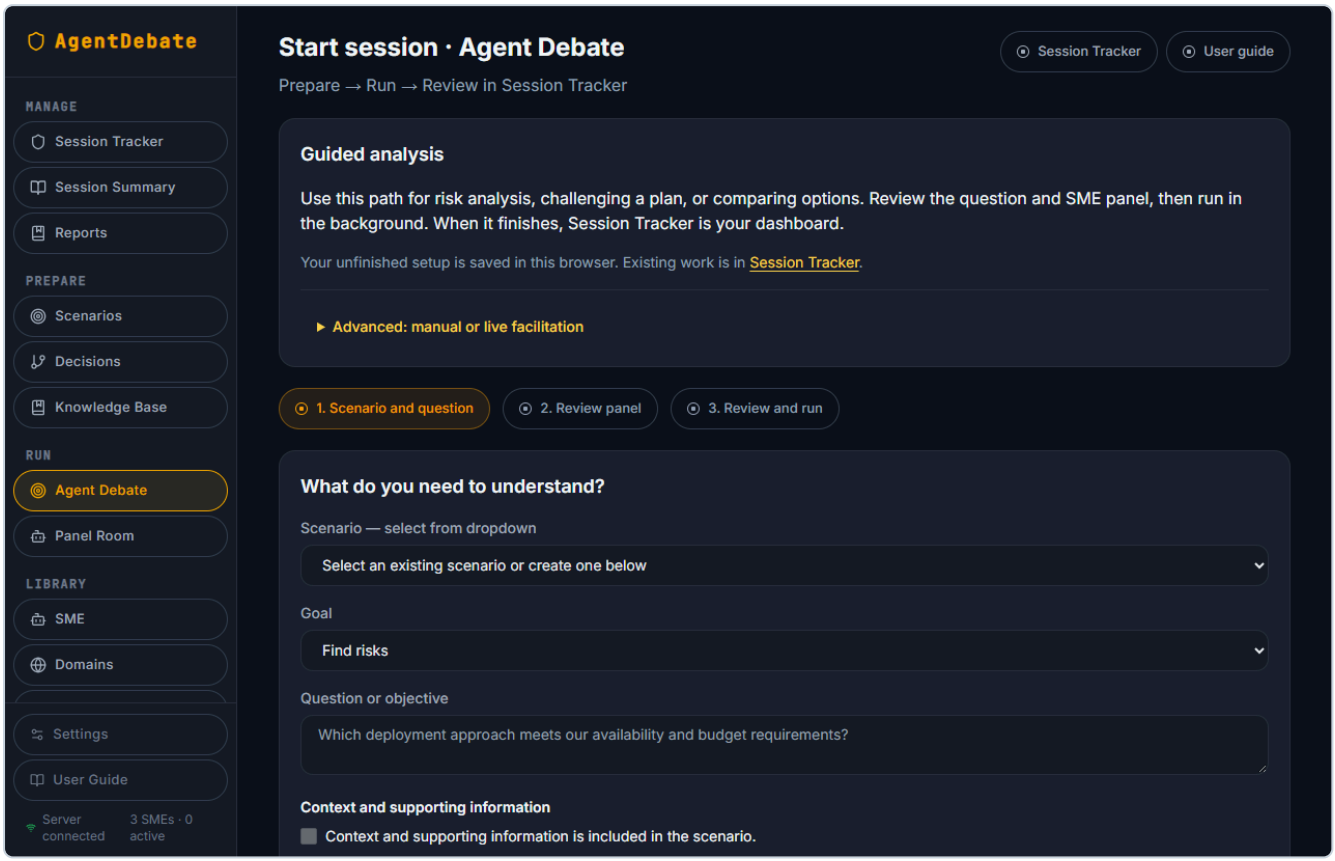


Figure 3. Guided setup begins with the intended analytical outcome. · Demonstration data

1. Open **Run** → **Agent Debate**. Select **Find risks**, **Challenge a plan**, or **Compare options**.
2. Enter the question and scenario. Check **Context and supporting information is included in the scenario** when the saved context is sufficient; otherwise enter the additional session context.
3. Review the suggested panel. For comparisons, provide 2–6 distinct options and acceptance criteria. Review optional assumptions, constraints, threats/chains, and sources.
4. Check readiness, then choose **Run analysis in background**. Return through Session Tracker to see status, inspect responses, and review the brief.

Automatic knowledge retrieval selects relevant excerpts; it does not send the entire library. An explicit empty source selection uses no knowledge documents. The worker saves completed SME work, synthesis, and resulting analysis material.

Set up Panel Room

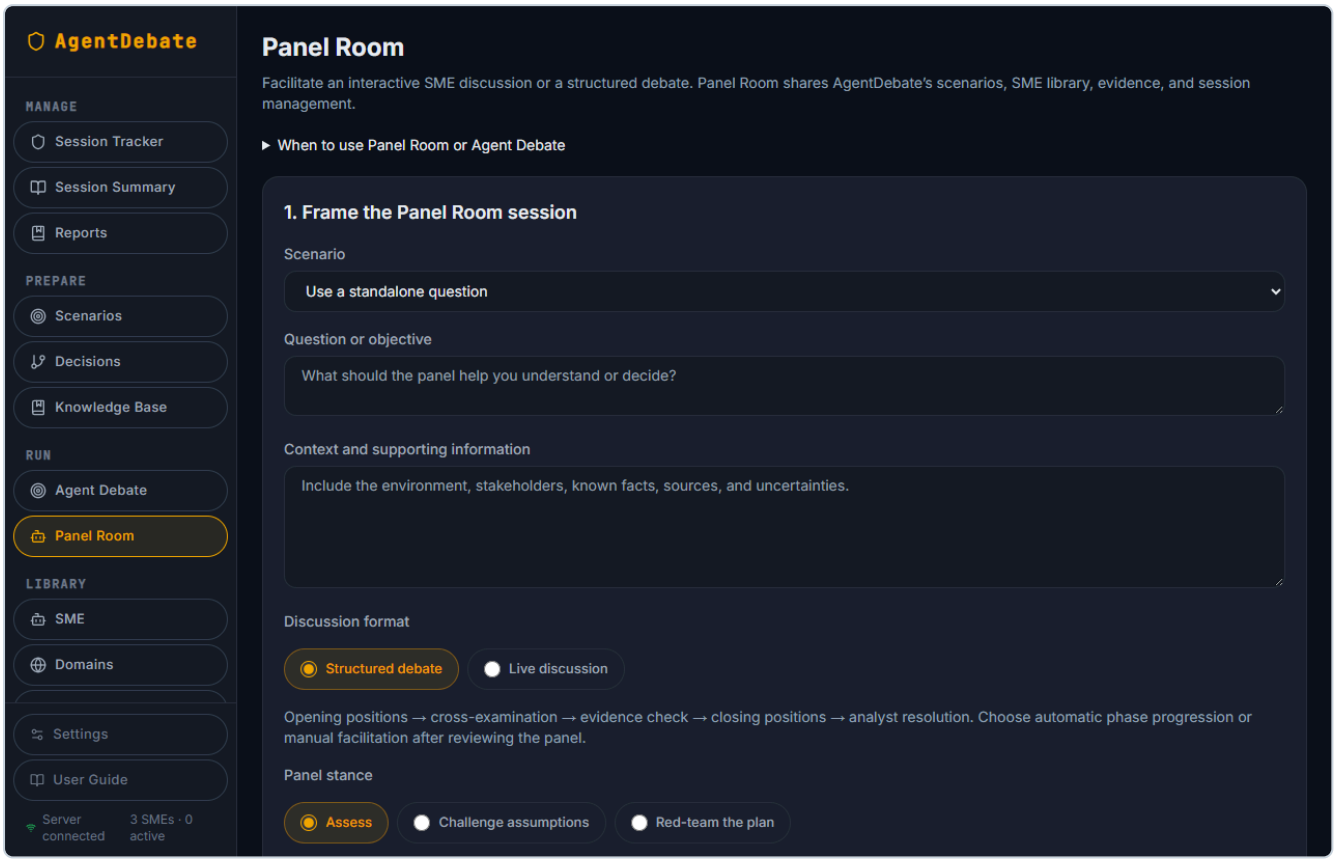


Figure 4. Panel Room combines session discovery and new debate setup. · Demonstration data

1. Open **Run → Panel Room**. Choose a scenario or frame a standalone question and supporting context.
2. Choose **Structured debate** for staged opening, challenge, evidence, and closing discussion; choose **Live discussion** for a discussion without fixed phases. Review the stance.
3. Choose **Review suggested panel**, then inspect the SMEs and context.
4. Choose **Start automatic debate** or **Open for manual facilitation**. Use the session list to return to saved work.

Panel Room snapshots the scenario context and SME profiles for a new session. Later scenario edits do not silently replace that starting context. Add relevant new facts or questions explicitly as analyst input.

> Automatic discussion stops for analyst resolution. It does not approve a decision or complete the session on your behalf.

Facilitate the structured debate

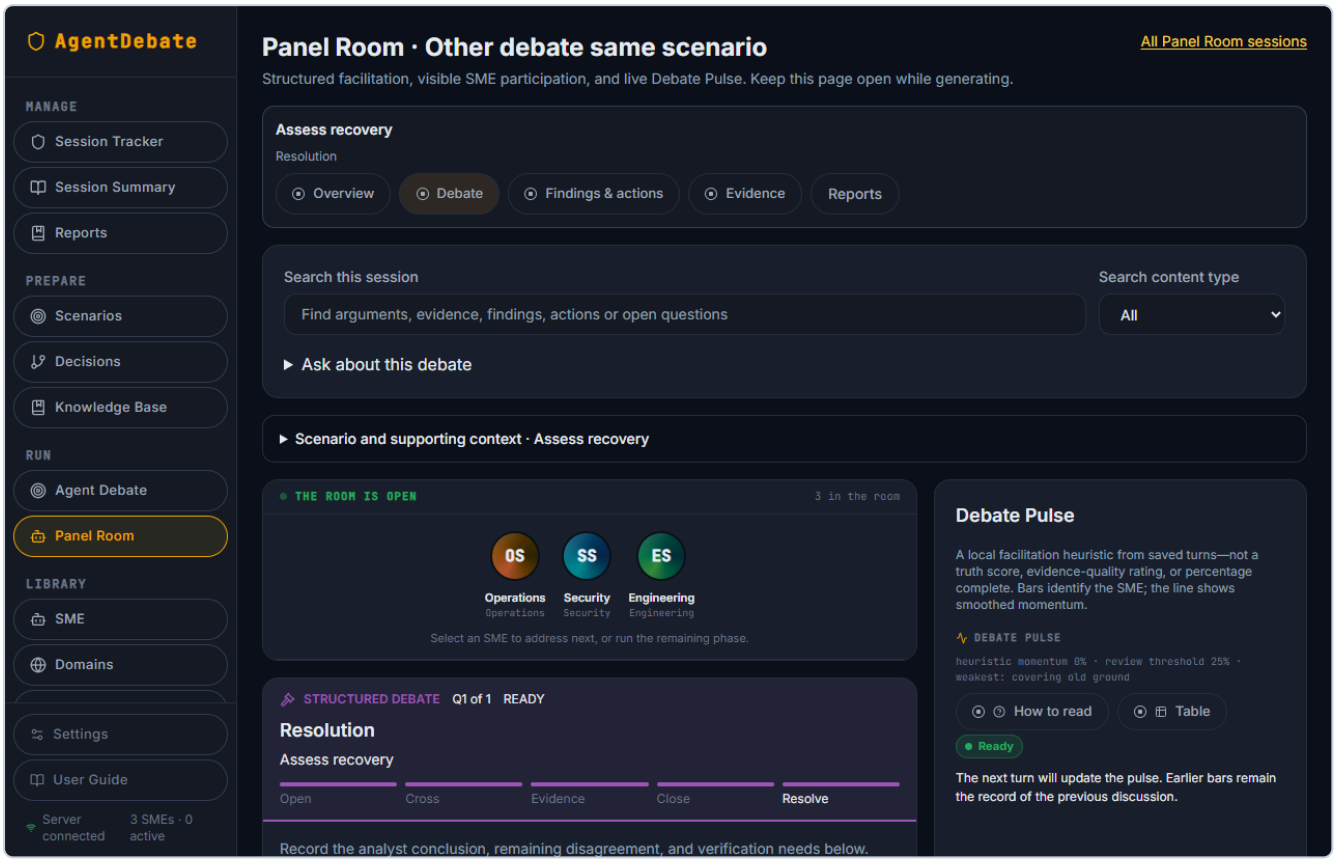


Figure 5. The debate workspace organizes speakers, phases, controls, and saved discussion. · Demonstration data

Phase	Analyst purpose
Opening positions	Establish each SME's initial interpretation and assumptions.
Cross-examination	Challenge claims, surface disagreements, and test alternative explanations.
Evidence check	Separate supported statements from gaps, assumptions, and model inference.
Closing positions	Identify revised positions and remaining disagreement.
Resolution	Record your conclusion or unresolved outcome, then complete the session.

Use the SME avatars and **Invite next SME contribution** for individual turns, or **Run remaining phase** for partial automation. **Run debate automatically** continues through the structured phases. Search, phase/SME filters, bookmarks, and Jump to latest help locate relevant contributions.

Read Debate Pulse and processing status

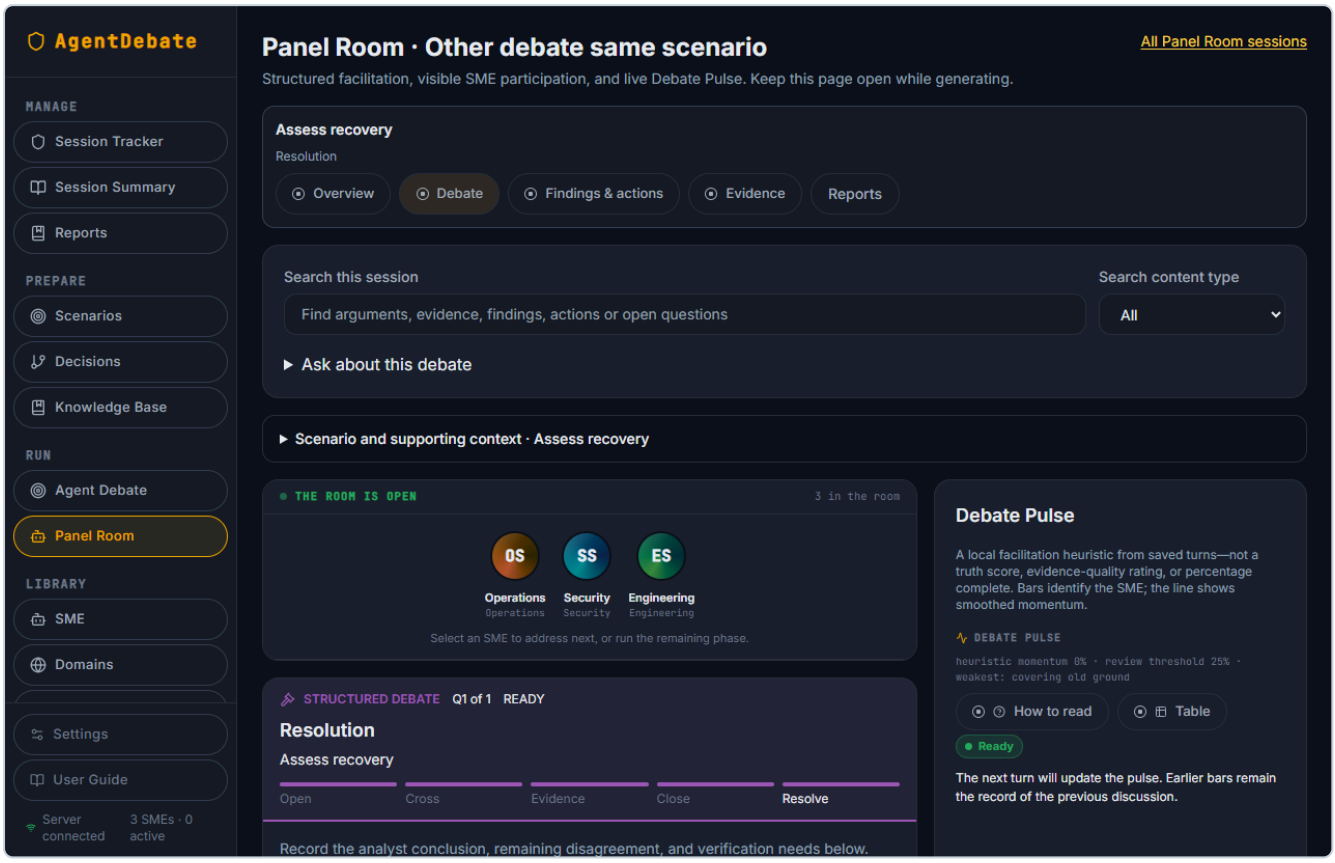


Figure 6. Debate Pulse and saved discussion support facilitation. · Demonstration data

Debate Pulse scores saved contributions using vocabulary novelty, engagement with other perspectives, and textual specificity. The bars identify speakers; the line smooths scores. Use its table and legend for precise interpretation. These are facilitation signals, not truth scores or time estimates.

Straw polls record model stances on an explicit proposition. Compare movement only across completed polls with the same proposition and roster. A settled poll does not establish correctness. Repetitive cross-examination can advance an automatic run to evidence check.

Processing feedback identifies the current effort. Report generation uses Prepare → Generate → Check → Save, with elapsed time, received text activity, or completed evidence-section counts when available. Streaming SME text is provisional until validated and saved.

> **Pause after current turn** saves that contribution before pausing. **Stop generation** cancels the current response and retains earlier saved work. Slow responses alone do not establish a connectivity or model failure.

Turn Panel Room discussion into results

1. After closing positions, open **Findings & actions** and choose **Generate results draft**.
2. Inspect each takeaway, disagreement, finding, threat, evidence-gap question, or proposed action. Follow the saved source references. Edit, accept, or dismiss each item deliberately.
3. Use **Inspect evidence / capture** on a contribution to capture a relevant passage. **Ask about this debate** provides model-assisted answers grounded in saved session references.
4. Accepted findings and threats become session-linked records needing review. Accepted actions need observable completion criteria; accepted questions feed the tracker's unresolved questions.
5. Record the analyst resolution. Use **Complete session and enable reports**, then continue the analytical review in Session Tracker.

What source validation actually does

The results workflow provides source and passage identifiers. The application resolves valid selections to exact saved text and checks references. Invalid source selection triggers a bounded correction attempt. If validation still fails, it does not replace previously saved results.

Matching a quotation proves that the cited text exists in the saved record. It does not prove that its claim is true. A quotation from an SME response remains model-generated material.

A useful action is testable

Weak action	Action with a completion condition
Investigate backup concerns.	Run the peak-load recovery test and record whether recovery meets the target.
Improve evidence.	Obtain the dated exercise log and document which failure stage caused the delay.

> Completing the debate, accepting a result, verifying a finding, and completing analyst review are separate events. None should be treated as a shortcut for the others.

Manage the session in Session Tracker

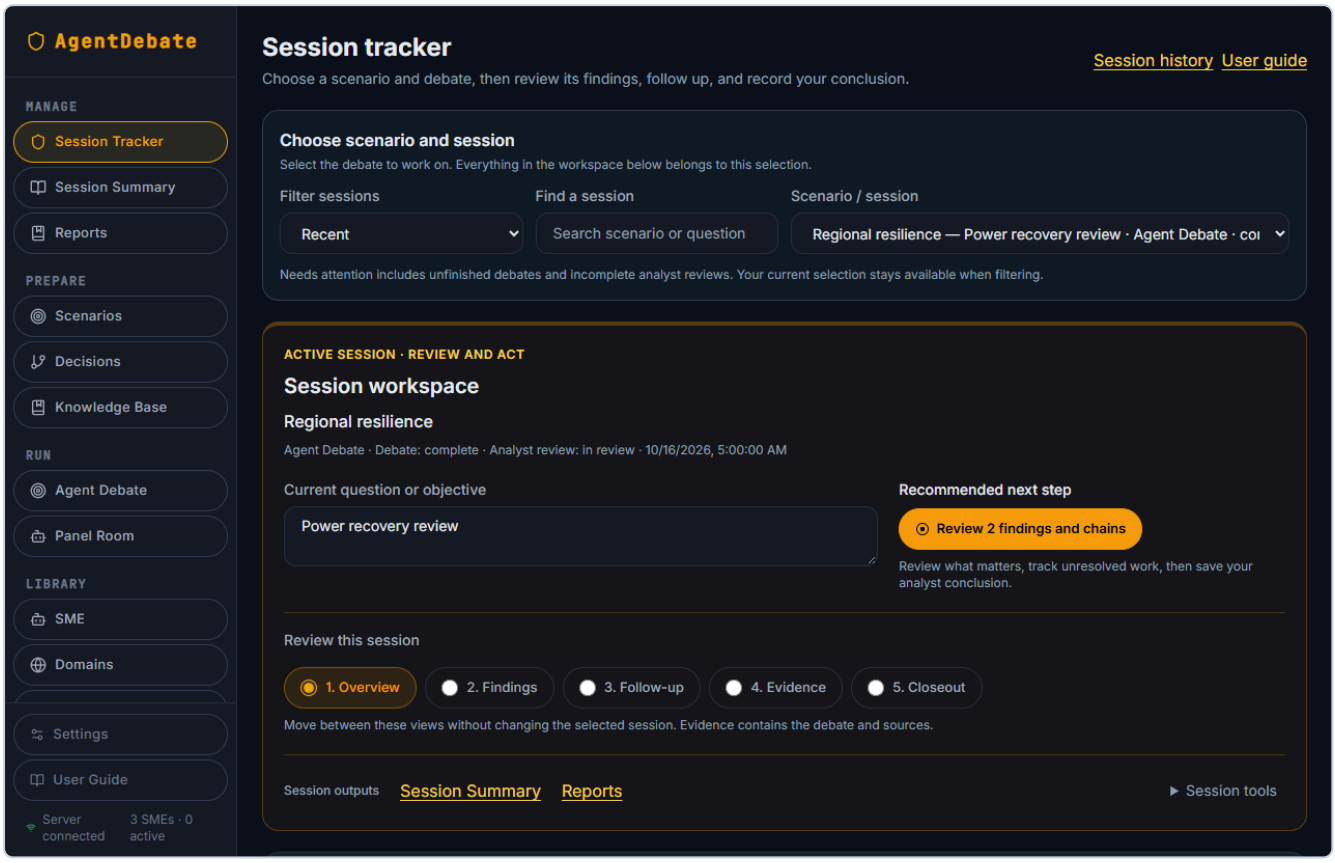


Figure 7. Selection and active workspace are deliberately separate. · Demonstration data

1. Use **Choose scenario and session** to select the debate. Search or filter by Recent, Needs attention, or Completed debates. Confirm the question, date, and mode.
2. In the amber-accented **Session workspace**, follow the recommended next step or choose Overview, Findings, Follow-up, Evidence, or Closeout.
3. Use **Search this session and compare debates** to retrieve saved material, including debate text. When an earlier completed debate is available, inspect what changed.
4. Prioritize unresolved findings by whether they could change the decision. Review the highest-impact uncertainty before polishing the narrative.

Session outputs contains Summary and Reports. **Session tools** contains maintenance/navigation controls such as refresh, run details, or returning to a completed Panel Room. Changing a review view does not change the selected session.

Verify findings and manage follow-up

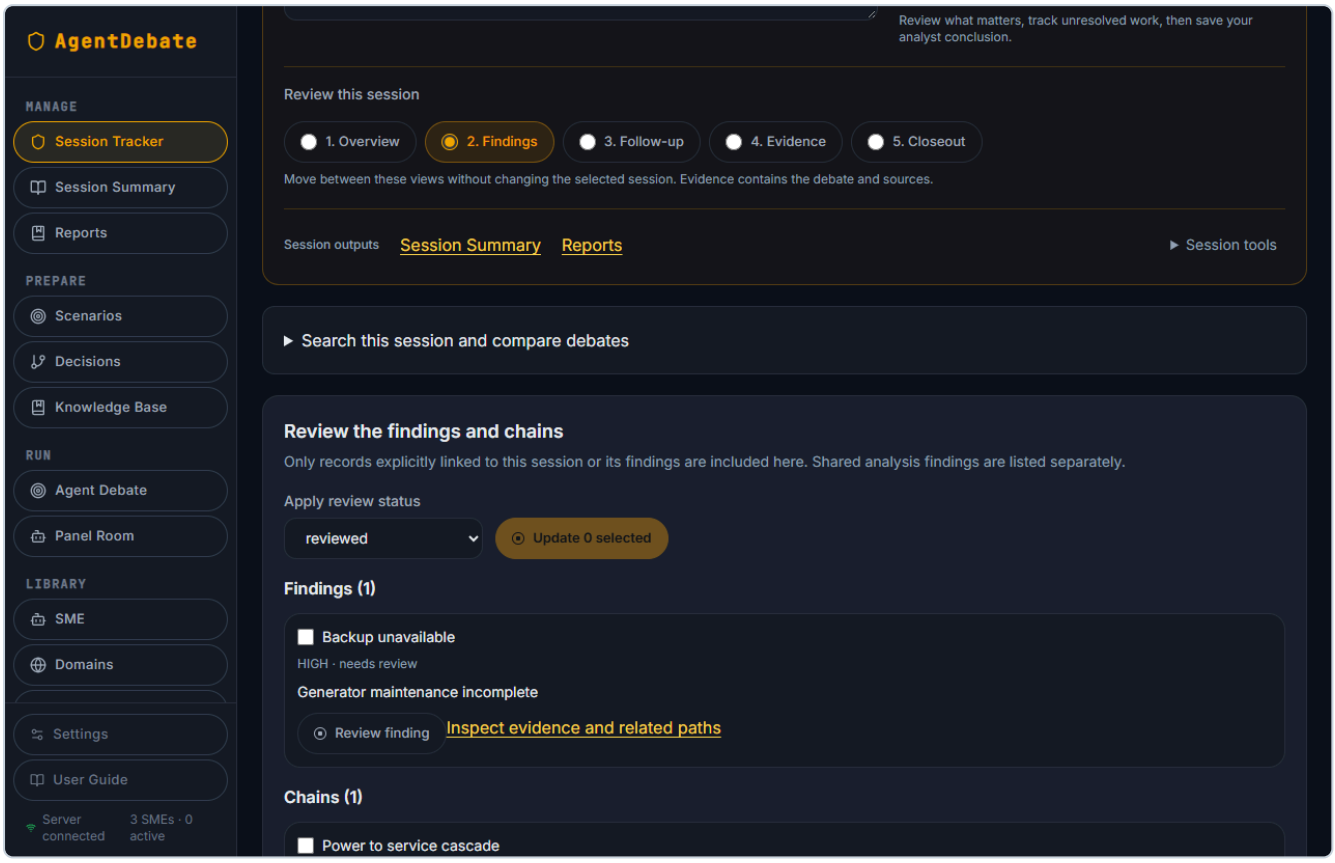


Figure 8. Findings review operates on records linked to the selected session. · Demonstration data

Open a finding review to assess evidence, confidence, impact on the decision, and linked work. Inspect exact supporting and opposing passages. Source age, relevance, quality, and independence matter more than simulated agreement.

Model inference means no exact supporting quotation was verified for that finding. It is a prompt to seek evidence, not a declaration that the claim is false. Automated support ratings are not probabilities of truth.

Use verification tools to check new or changed claims. Add source text or an allowed source URL, assess the retrieved passage, and save your judgment and rationale. Exported analyst-reviewed examples are local JSONL; exporting does not train a model.

In **Follow-up**, state the action, completion criterion, relevant finding/chain, and due date. Implemented work remains open until verified or rejected. A mitigation does not automatically reduce severity; reassess the underlying risk after evidence arrives.

Close out, summarize, and report

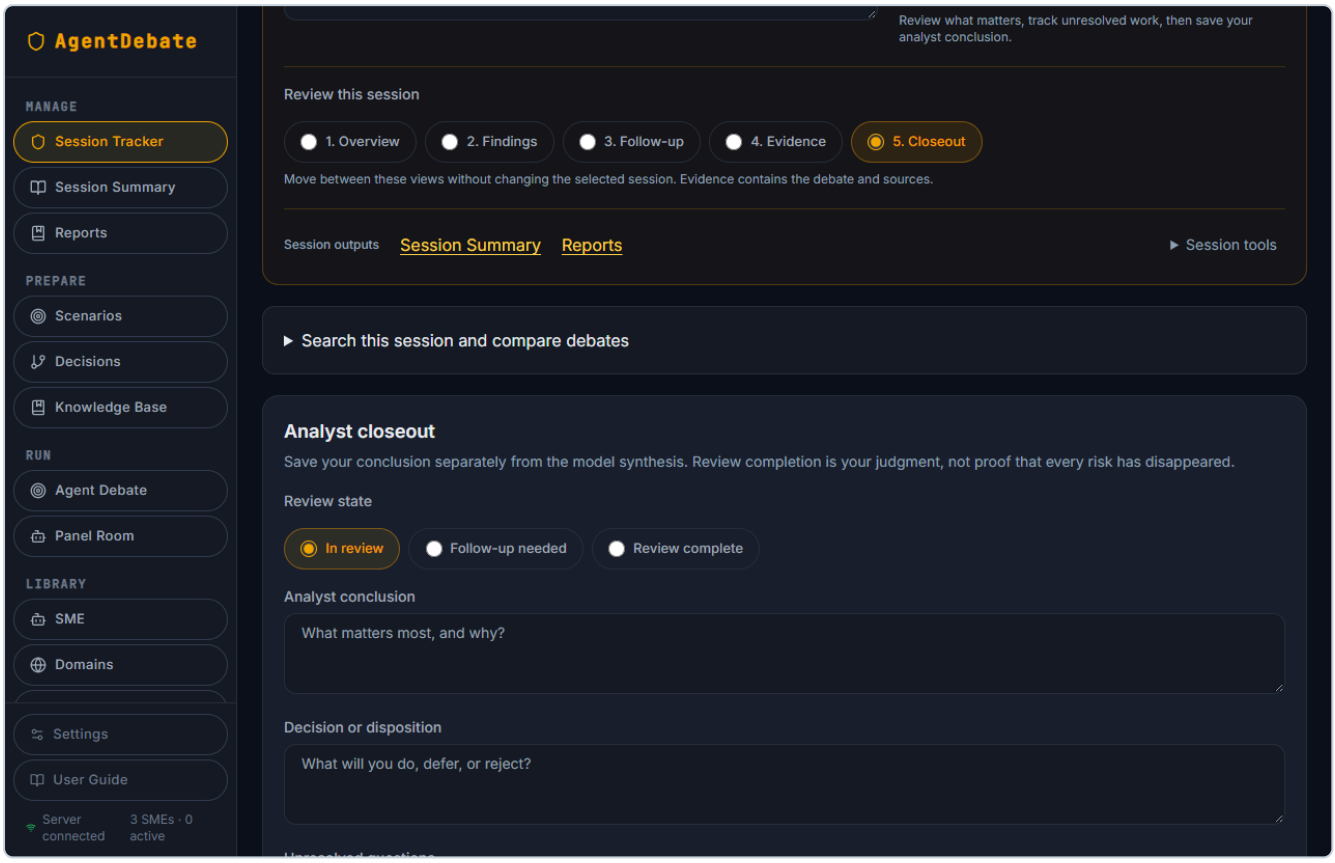


Figure 9. Closeout records the analyst's disposition separately from model output. · Demonstration data

1. Record your conclusion, decision or disposition, unresolved questions, revisit conditions, and accepted risks or deferred work.
2. Notes autosave as a draft. Choose **Save session review** to confirm the review state; autosave alone does not finish the review.
3. To mark review complete, finish the debate and provide a conclusion and decision. If work remains, explain and acknowledge it.
4. Use **Session Summary** for the scenario, conclusion, findings, threats, mitigations, and other session information in one reference view.
5. Use **Reports** to produce a specific shareable artifact and inspect it before export.

> New evidence or changed session material may require reassessment. A saved review is an analyst judgment at a point in time, not permanent verification.

Choose and export a report

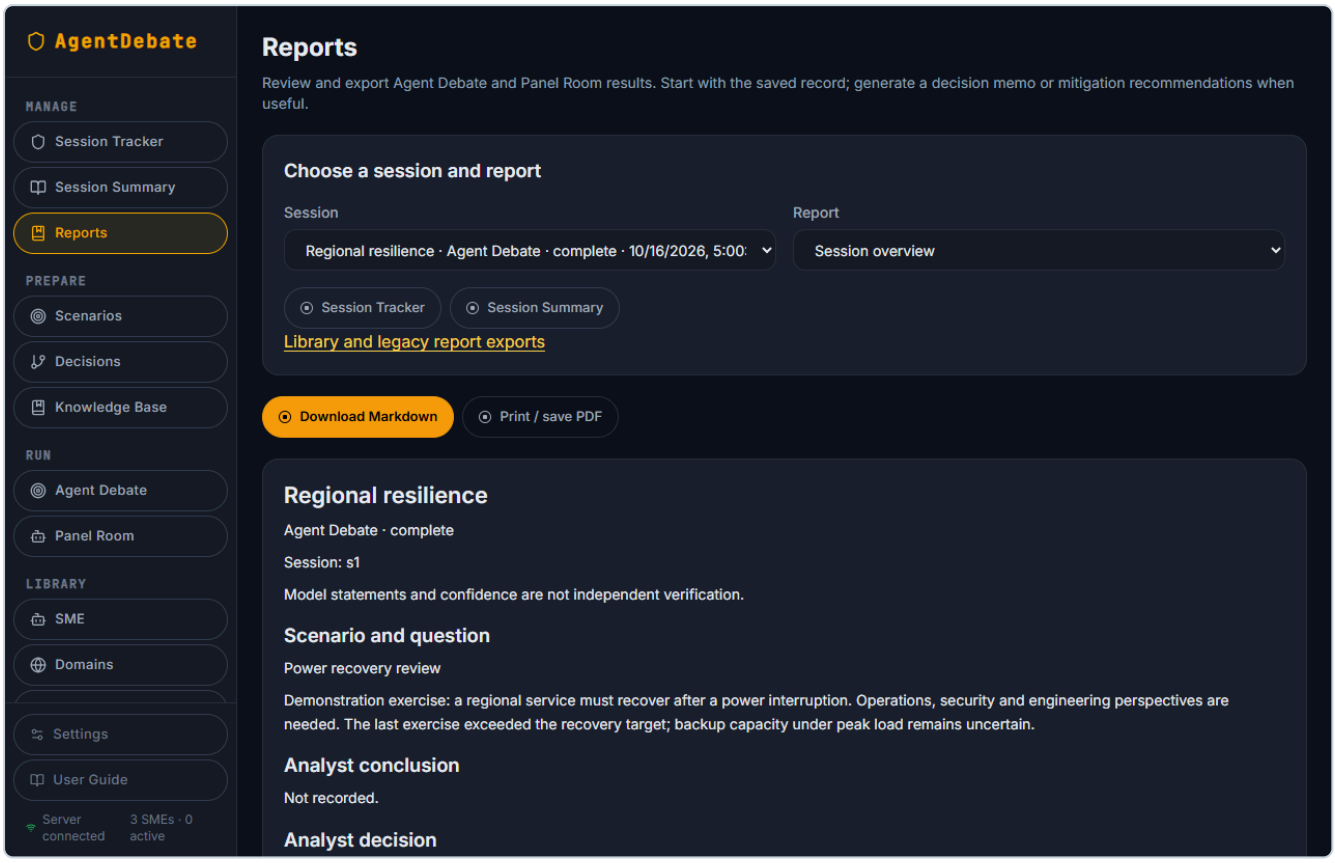


Figure 10. Reports selects a session and an output type. · Demonstration data

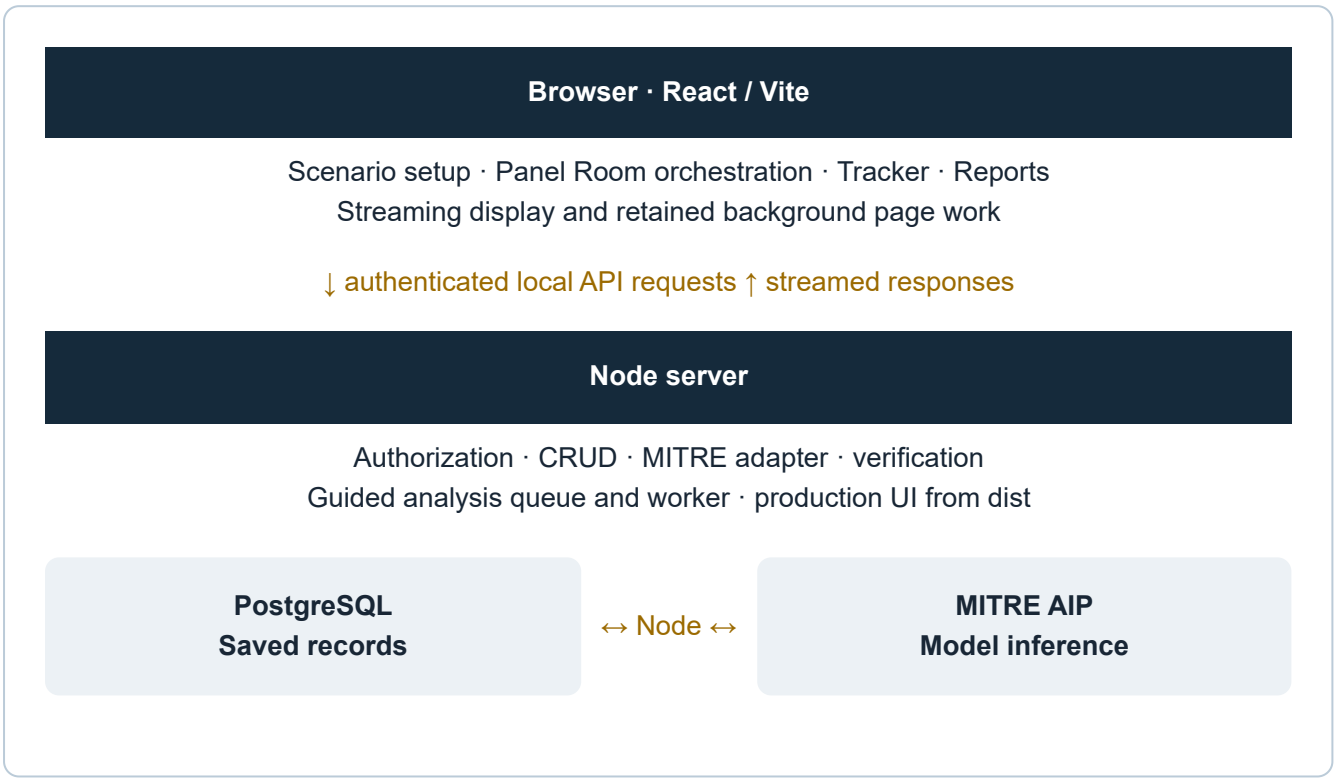
Report family	How to use it
Saved-record reports	Session overview, full debate, SME positions, evidence/source context, Pulse/polls, and Panel Room reviewed results organize saved records without requiring a new model narrative.
AI-assisted reports	Decision memo and mitigation/follow-up recommendations use model generation after session completion. Treat the result as a draft; check cited evidence and unresolved dissent.

For long records, generation can review evidence in sections before combining results. Follow the current processing stage. A source/quote validation failure leaves the prior saved report unchanged; it does not authorize an unsupported final report.

Use **Mark report reviewed** after your inspection, then **Download Markdown** or **Print / save PDF**. Source changes can make a draft outdated. Report review does not verify every underlying claim or create mitigation records automatically.

Implementation: data and model flow

The local application combines a React interface, a Node API and worker, PostgreSQL storage, and the MITRE AIP model service.



Work	Persistence and lifetime
Guided Agent Debate	Server worker continues independently of browser navigation or closure while the server remains running. Saved progress supports recovery.
Panel Room and browser report generation	Continue across application pages through retained active views. Keep the same browser tab open. Reload, sign-out, or closure interrupts in-flight browser work.
Panel contributions	Transcript, protocol state, and position updates are saved together in the contribution record. A save failure stops progression.
Analyst drafts and outputs	Saved to the local database; tracker notes also use a browser recovery copy for pending edits.

The AIP bearer key stays on the server. The browser calls the local API; Node translates requests to the configured AIP endpoint. Local storage does not mean offline inference: selected prompt context is sent to MITRE AIP.

Implementation: prompts and evidence

Stage	Information assembled into the prompt
SME assessment	Question, scenario context, selected SME profile, constraints/assumptions, relevant knowledge excerpts, and selected analytical material.
Rebuttal or subsequent Panel Room turn	Phase instructions, current question, selected prior remarks, recent discussion, saved positions, and relevant source passages.
Synthesis / comparison	Saved SME contributions, disagreements, options and criteria where applicable, and evidence needed to form a brief.
Results / reports	Saved session records with source references; section review when the record is too large for a single synthesis.
Verification	The claim and candidate source passages, with supporting/opposing evidence and limitations to assess.

Preserving context without pretending it is unlimited

Knowledge retrieval selects passages, not whole libraries. Panel Room uses selective recall and recent context; the complete transcript remains saved even though every turn does not receive all earlier text.

Guided SME rounds request 4,096 output tokens; synthesis/comparison briefs request 6,144. The MITRE adapter can retry output-limit failures up to twice with a larger allowance, bounded by the configured ceiling (8,192 by default). Incomplete output is not treated as a completed result.

Section review uses quotation and coverage checks. Those checks help preserve provenance; they do not guarantee semantic equivalence to every original statement. Return to the original evidence for consequential decisions.

Where the implementation lives

`server/analysisQueue.js` and `analysisWorker.js`: guided processing.
`server/llmProvider.js`: provider translation and recovery.
`src/lib/panelWorkflow.js` and `src/lib/panel/`: structured turns, phases, and positions.
`src/lib/panelReview.js`: results and reference validation.
`server/verificationService.js`: finding verification.
`src/components/layout/BackgroundWork.jsx`: retained active page work.

Operate the local installation

Existing Windows installation

From the `AgentDebate-local` repository in PowerShell:

```
./scripts/start-native-local.ps1 -RepositoryPath .
```

Open `http://localhost:3000`. The documented native setup uses PostgreSQL on `127.0.0.1:55432`, with binaries, database files, and logs under `%LOCALAPPDATA%AgentDebate-local`. This keeps database data outside the OneDrive repository.

Configuration and build

Setting / command	Purpose
DATABASE_URL	Connection to the intended PostgreSQL database; preserve the native port when using the existing setup.
LLM_PROVIDER=mitre_aip	Select the MITRE adapter explicitly.
MITRE_AIP_BASE_URL / API_KEY / MODEL	Server endpoint, secret, and accessible default model. Keep the ignored .env out of exports and Git.
HOST / PORT / APP_ORIGIN	Default local listener and browser origin. Local binding is not organization-wide hosting.
npm ci → npm run build → npm start	Install locked dependencies, build the UI, and launch Node. Requires Node 22+ and configured PostgreSQL.
npm run db:migrate	Explicit migration command. Server startup also applies pending migrations.
npm run verify	Repository lint, tests, and production build; mock/provider tests are not a live AIP health check.

After frontend changes, rebuild `dist` and refresh the browser. After server or environment changes, restart the application process. Do not overwrite an existing `.env` with example values.

In **Settings**, refresh the accessible model list and test the connection before saving a model. Node needs network/VPN access and appropriate CA trust. Do not disable TLS verification to bypass a certificate failure.

> For MITRE-wide availability, plan an approved host, HTTPS, access controls, network reachability, backups, and an operating owner. The repository includes Docker configuration, but this guide does not certify that deployment as tested or approved.

Troubleshooting and recovery

Symptom	Interpretation and next step
SME is “gathering thoughts”	Read the current connection/waiting/streaming stage and elapsed time. Incoming text indicates activity. A healthy local server does not establish upstream AIP health.
Slow report generation	Check section counts and Generate / Check / Save stages. Large evidence reviews and bounded corrections take additional calls; the bar is not a reliable time estimate.
Output token limit	The response ended before completion. Allow bounded recovery; if it still fails, inspect model/configuration limits. Do not accept truncated output as final.
Malformed JSON	The model response did not meet the required structure. The Panel Room path handles trailing commas and one correction retry; persistent failure stops the turn.
Unmatched source / quotation	A reference failed validation. Inspect saved source text and retry after correcting the source issue. Prior saved results remain available.
Contribution save failed	Retry saving the unsaved contribution before resuming the debate. Navigating away is different from reloading or closing the browser.
Stopped or interrupted automatic debate	Return to the session and resume automatic debate from saved progress. Each contribution has a 15-minute timeout; there is no one-hour total debate cutoff.
Tracker draft save failed	Use Retry draft save. Preserve pending browser text until database saving succeeds.
Local page unavailable	Check the application process and app logs, then PostgreSQL. Start the documented local script when processes are stopped.

Operational recovery

Inspect `app.stdout.log`, `app.stderr.log`, and `postgres.log` under the local runtime directory. Confirm the recorded PID belongs to this application before stopping a process. Do not delete the database directory to resolve an application error.

Recommended operational practice: back up PostgreSQL with the organization’s approved tooling and test restoration into a separate database. A Git copy contains application code, not the running database contents or ignored credentials.

Analyst quick reference

Before running

1. State a decision-relevant question. Identify the scenario, facts, assumptions, and missing evidence.
2. Choose Agent Debate for guided analysis or Panel Room for facilitated/structured discussion.
3. Review the SME panel, sources, and any option criteria.

Before accepting a finding

1. Read the claim and its exact source passage. Determine whether the source is supplied evidence or model output.
2. Check opposing evidence, independence, age, and relevance. Record the limitations.
3. Assess whether resolving the uncertainty could change the decision. Create a testable follow-up when needed.

Before sharing

1. Record your conclusion, disposition, accepted risks, and conditions for revisiting.
2. Confirm the selected scenario/session and whether the report reflects the latest saved record.
3. Review the report and export. Do not imply that model agreement or a matched quotation establishes truth.

Implementation references in the repository

`src/pages/UserGuide.jsx` — current in-application workflow guidance.
`docs/LOCAL_RUNNING.md` — this Windows runtime.
`docs/MITRE_AIP_SETUP.md` — provider configuration.
`docs/SESSION_TRACKER.md` and `SESSION_SUMMARY.md` — analyst management.
`docs/VERIFICATION_AND_REVIEWED_EXAMPLES.md` — review and evaluation examples.
`docs/PANEL_ROOM_INTEGRATION.md` — integration background.

Prepared from local source and current UI on 22 October 2026. Current implementation and in-app guidance were preferred where older documents differed, including automatic-run lifetime and result passage validation. Screenshots use simulated records and model responses. No live AIP health or hosting approval is claimed.